

MH371 truth-blind cruise control

conditional_antenna_full_precomp_medium_bfo | 2 deterministic seeds | generated from run artifacts

Control result

MODEL FAMILY	PARTICLES
conditional_antenna_full_precomp_medium_bfo	12000 x 2 seeds
TERMINAL MEAN ERROR	TERMINAL MASS WITHIN 100 NM
21.8 to 30.9 NM	57.2 to 63.6%
TERMINAL ROOT ESS	BFO OBSERVATION SD
27.9 to 33.2	4.0 Hz
TRUTH-IMPLIED BFO BIAS	BTO EQUATION CLOSURE
167.2 to 171.6 Hz	3.076e-1 us max

Evidence and inference conditions

- Cruise only: 01:48 UTC initial state through the selected 06:48:33 UTC SATCOM observation.
- Later ACARS positions, trajectory, Mach, altitude, and headings were excluded from inference and loaded only here.
- ERA5 temperature and wind plus IGRF-14 east-positive magnetic declination drive every propagation step.
- Satellite oscillator and Perth GES AFC terms remain separate inputs; aircraft BFO bias is analytically estimated.
- Each map combines deterministic seed replicates with equal replicate weight.
- Conditional RxGain endpoint: full target-EIRP precompensation; event-level SD 1.72 dB; reference gain 12.16 dBic.
- Power rows use the selected event/channel aggregation; maximum absolute offset from its SATCOM arc is 73.922 s.

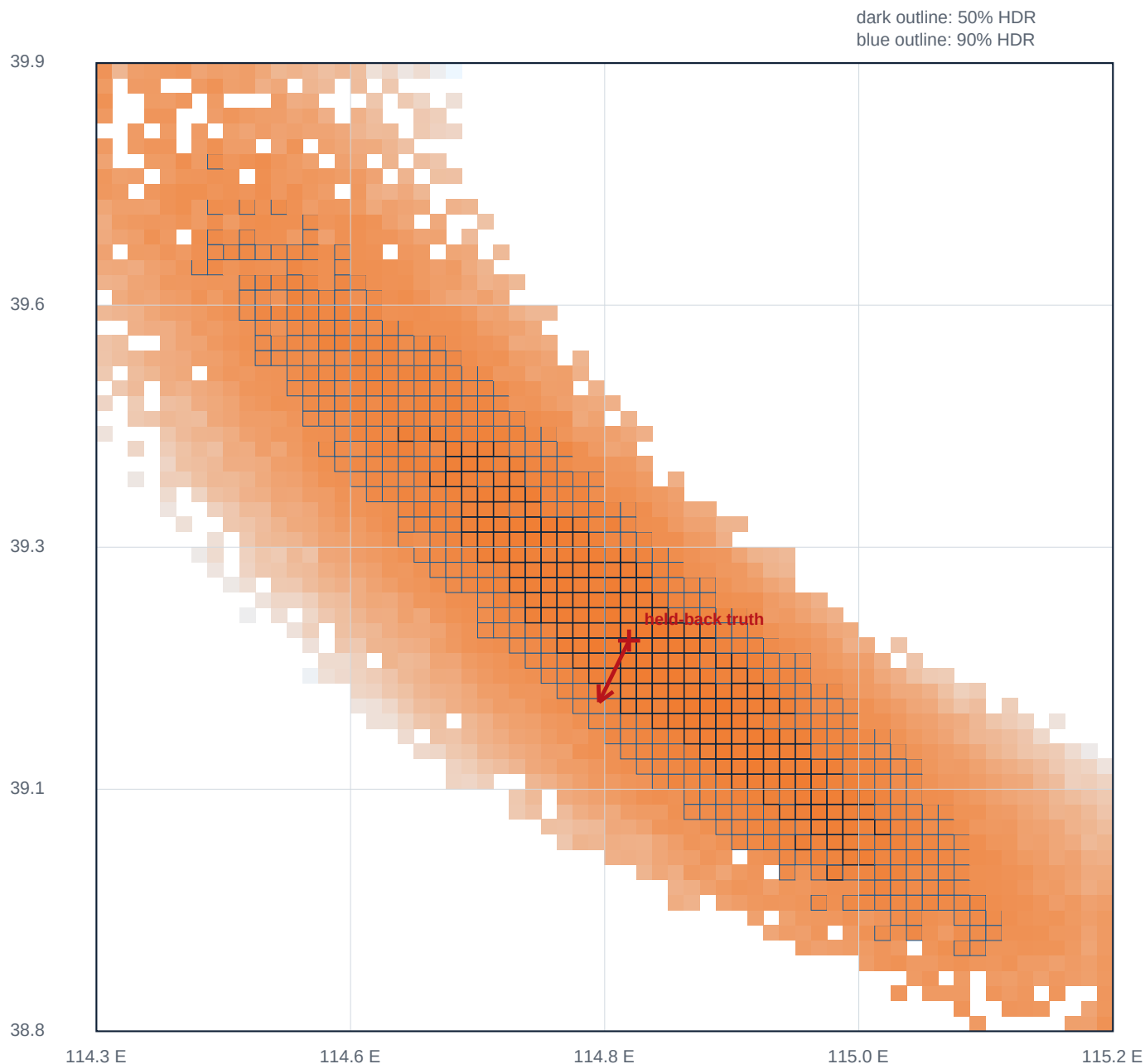
Material limitations

- held-back ACARS positions and trajectories were loaded only by this scorer
- posterior localization is conditional on the declared motion and BFO-noise family
- this known-flight control does not establish adequacy for the MH370 accident flight
- conditional antenna sensitivity only: RxGain validation did not justify unconditional core use or a model-family mixture weight
- the explicitly permitted first-pass gain surface is uncommissioned, lacks verified port/starboard handoff, and does not reproduce every workbook event gain
- the event/channel full-precompensation prediction is frozen; the antenna likelihood contributes only the modeled directional departure

The following 6 pages show posterior density, held-back position, and true-heading arrow at each selected SATCOM arc.

mh371_0200 SATCOM arc

2014-03-07T01:56:47.921Z | truth 39.2204 deg, 114.7725 deg | true heading 206.0 deg



Posterior and control metrics

MEAN POSITION ERROR

2.0 NM

MACH: POSTERIOR / TRUTH

0.783 / 0.759

GROUND SPEED: POSTERIOR / TRUTH

388 / 383 kt

TRUTH-IMPLIED BFO BIAS

167.2 Hz

POSTERIOR WITHIN 100 NM

100.0%

ALTITUDE: POSTERIOR / TRUTH

34263 / 27599 ft

LATENT BFO BIAS

166.9 Hz

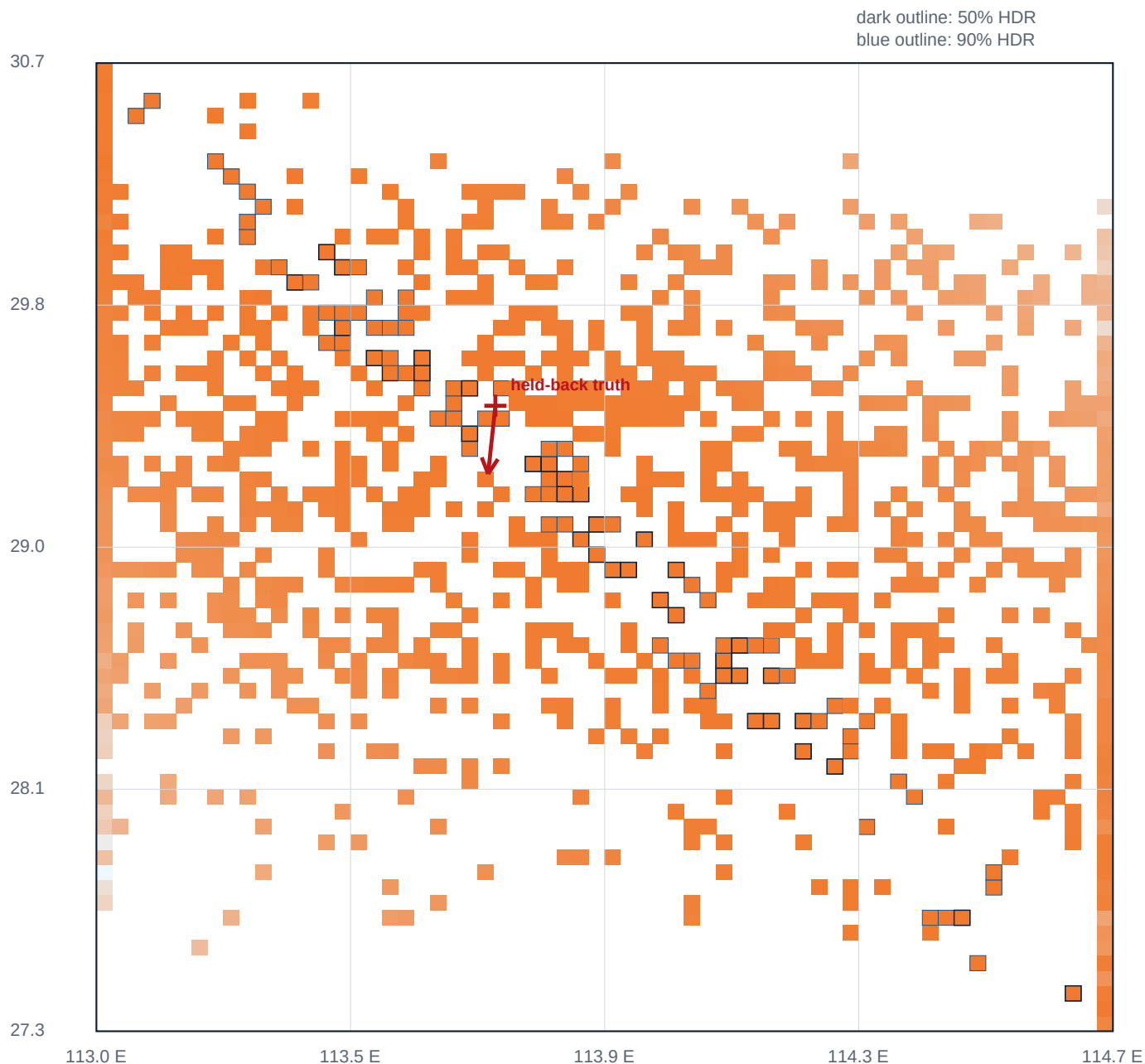
PARTICLE / ROOT ESS

7782 / 7782.4

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0300 SATCOM arc

2014-03-07T03:20:22.428Z | truth 29.4730 deg, 113.6856 deg | true heading 186.1 deg



Posterior and control metrics

MEAN POSITION ERROR

26.2 NM

MACH: POSTERIOR / TRUTH

0.787 / 0.753

GROUND SPEED: POSTERIOR / TRUTH

438 / 452 kt

TRUTH-IMPLIED BFO BIAS

171.6 Hz

POSTERIOR WITHIN 100 NM

94.9%

ALTITUDE: POSTERIOR / TRUTH

34317 / 28368 ft

LATENT BFO BIAS

168.6 Hz

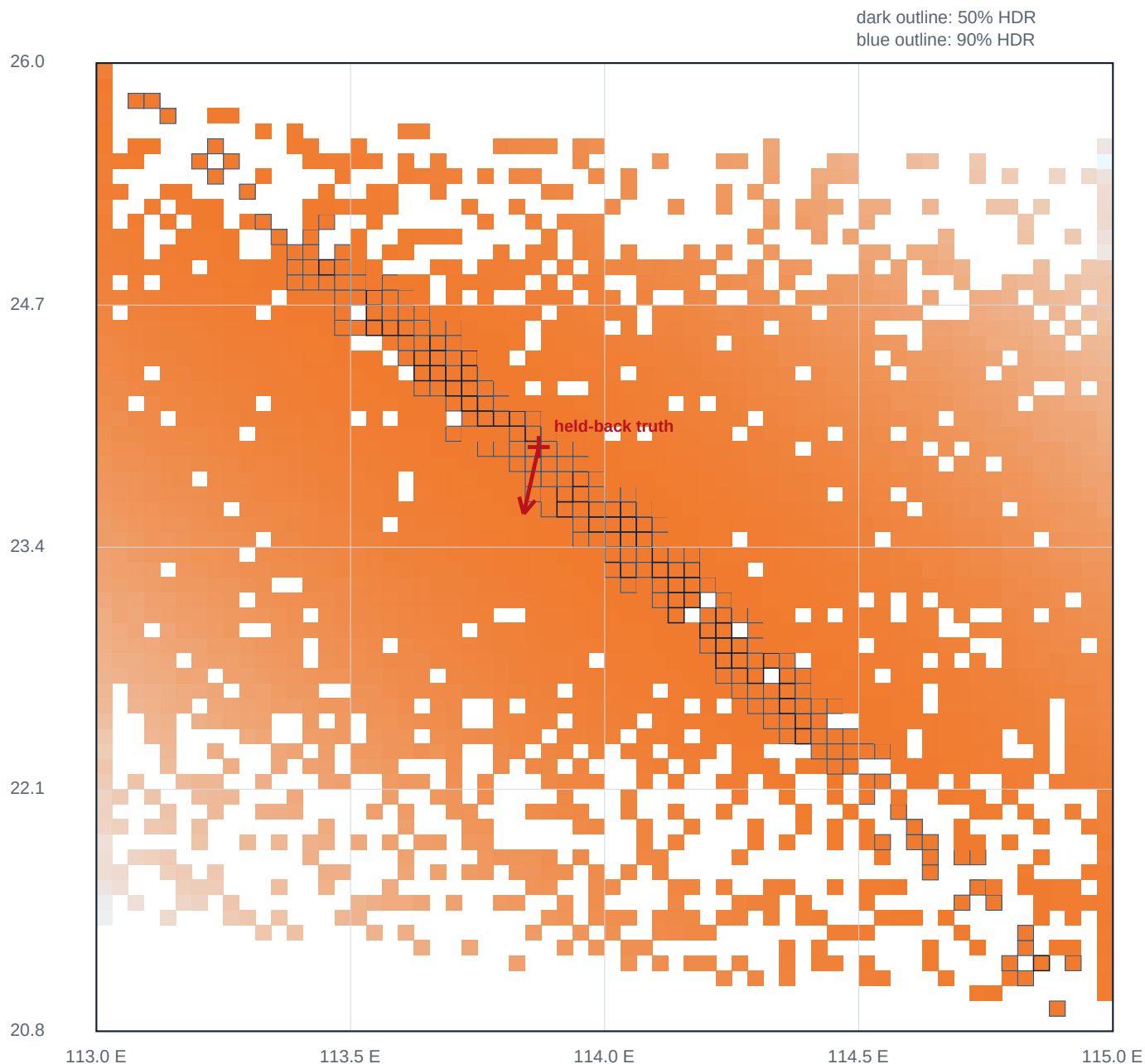
PARTICLE / ROOT ESS

72 / 72.1

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0400 SATCOM arc

2014-03-07T04:04:57.916Z | truth 23.9533 deg, 113.8817 deg | true heading 192.7 deg



Posterior and control metrics

MEAN POSITION ERROR

25.6 NM

MACH: POSTERIOR / TRUTH

0.786 / 0.832

GROUND SPEED: POSTERIOR / TRUTH

445 / 455 kt

TRUTH-IMPLIED BFO BIAS

169.6 Hz

POSTERIOR WITHIN 100 NM

89.4%

ALTITUDE: POSTERIOR / TRUTH

34393 / 38100 ft

LATENT BFO BIAS

168.9 Hz

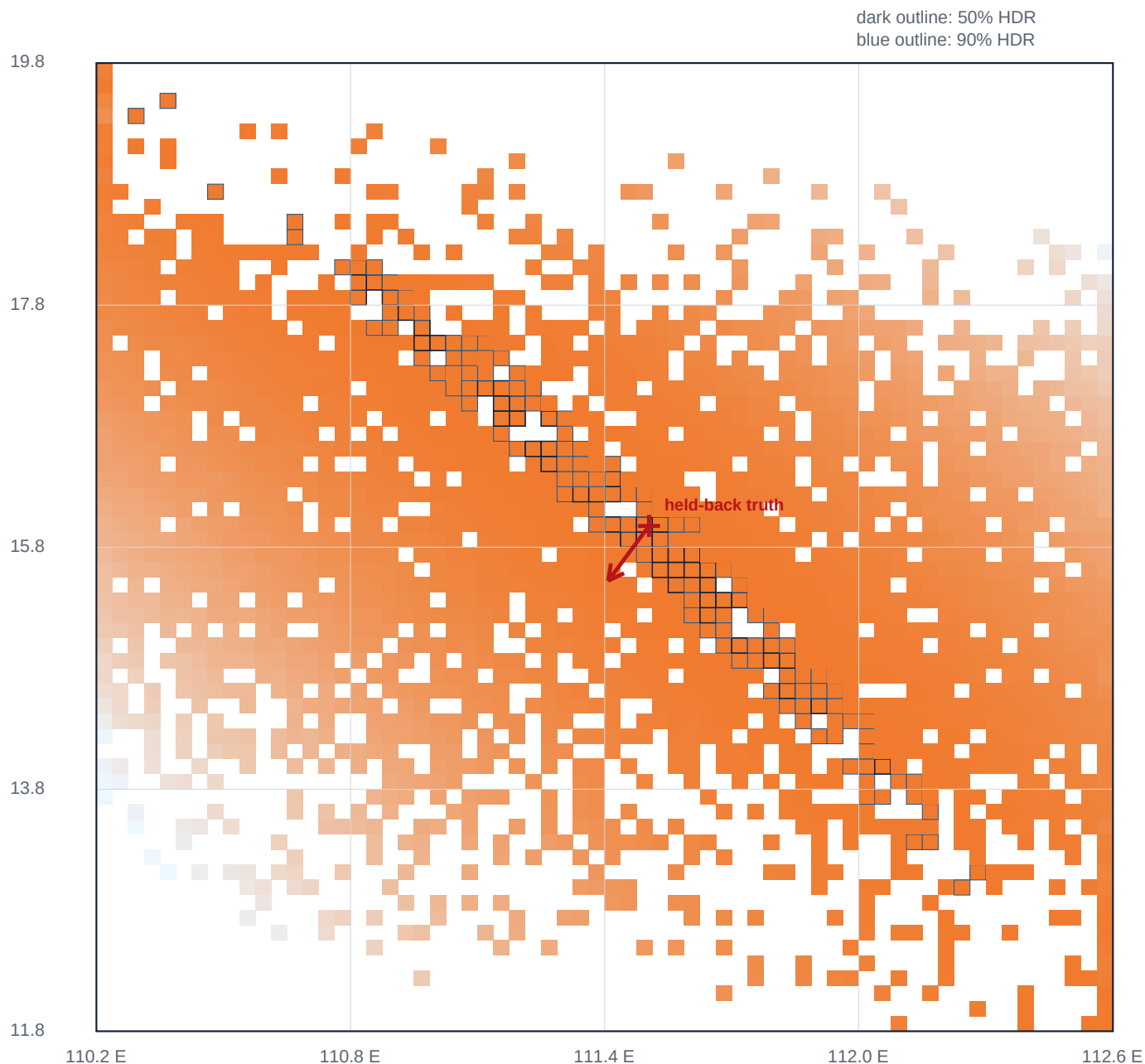
PARTICLE / ROOT ESS

295 / 61.1

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0500 SATCOM arc

2014-03-07T05:11:37.411Z | truth 15.9956 deg, 111.5330 deg | true heading 216.7 deg



Posterior and control metrics

MEAN POSITION ERROR

16.8 NM

MACH: POSTERIOR / TRUTH

0.777 / 0.830

GROUND SPEED: POSTERIOR / TRUTH

447 / 462 kt

TRUTH-IMPLIED BFO BIAS

169.7 Hz

POSTERIOR WITHIN 100 NM

76.5%

ALTITUDE: POSTERIOR / TRUTH

35216 / 39998 ft

LATENT BFO BIAS

168.7 Hz

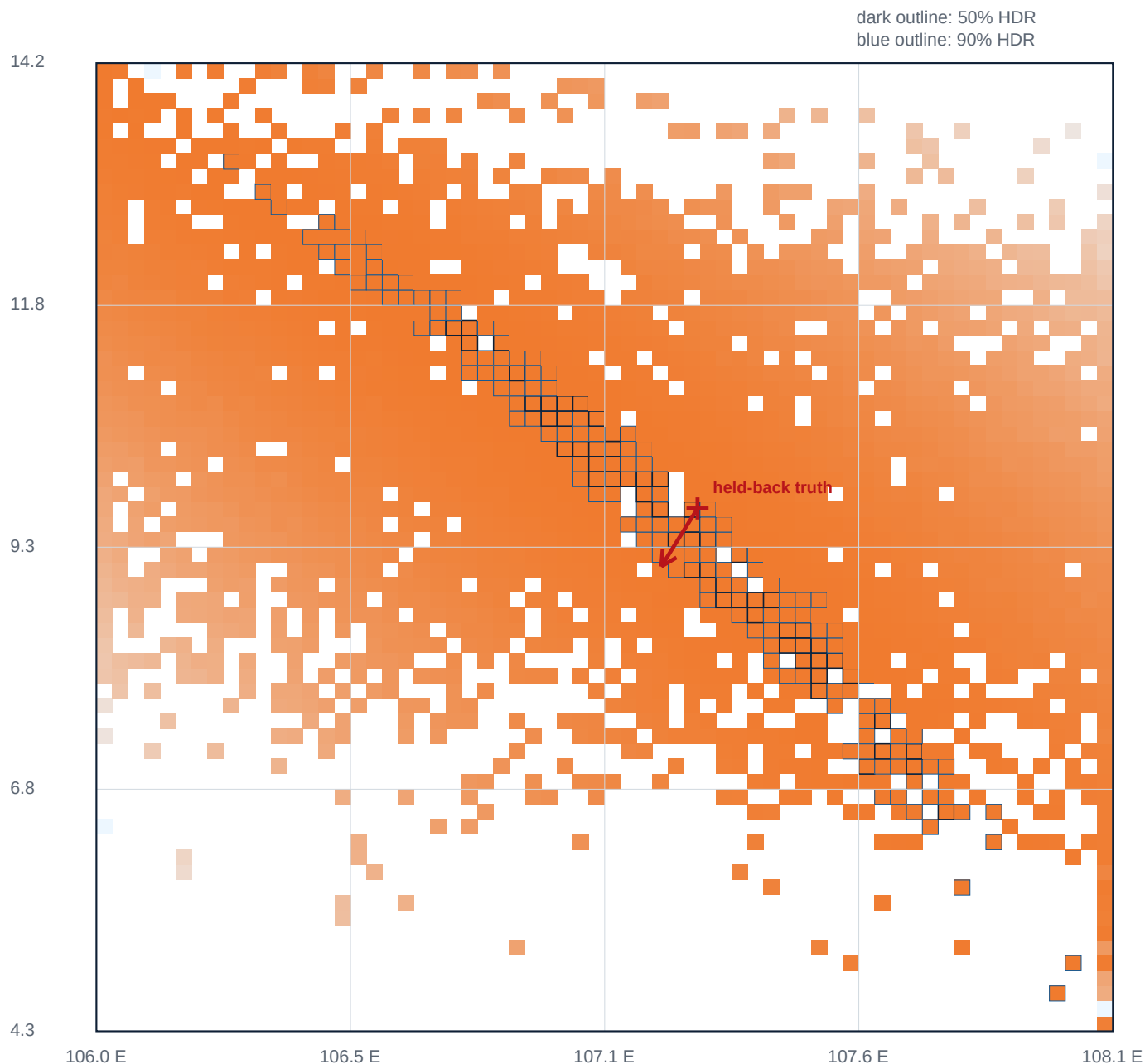
PARTICLE / ROOT ESS

167 / 44.8

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0600 SATCOM arc

2014-03-07T06:09:44.409Z | truth 9.6835 deg, 107.2620 deg | true heading 212.0 deg



Posterior and control metrics

MEAN POSITION ERROR

22.3 NM

MACH: POSTERIOR / TRUTH

0.788 / 0.813

GROUND SPEED: POSTERIOR / TRUTH

472 / 474 kt

TRUTH-IMPLIED BFO BIAS

170.6 Hz

POSTERIOR WITHIN 100 NM

65.4%

ALTITUDE: POSTERIOR / TRUTH

33917 / 39997 ft

LATENT BFO BIAS

169.0 Hz

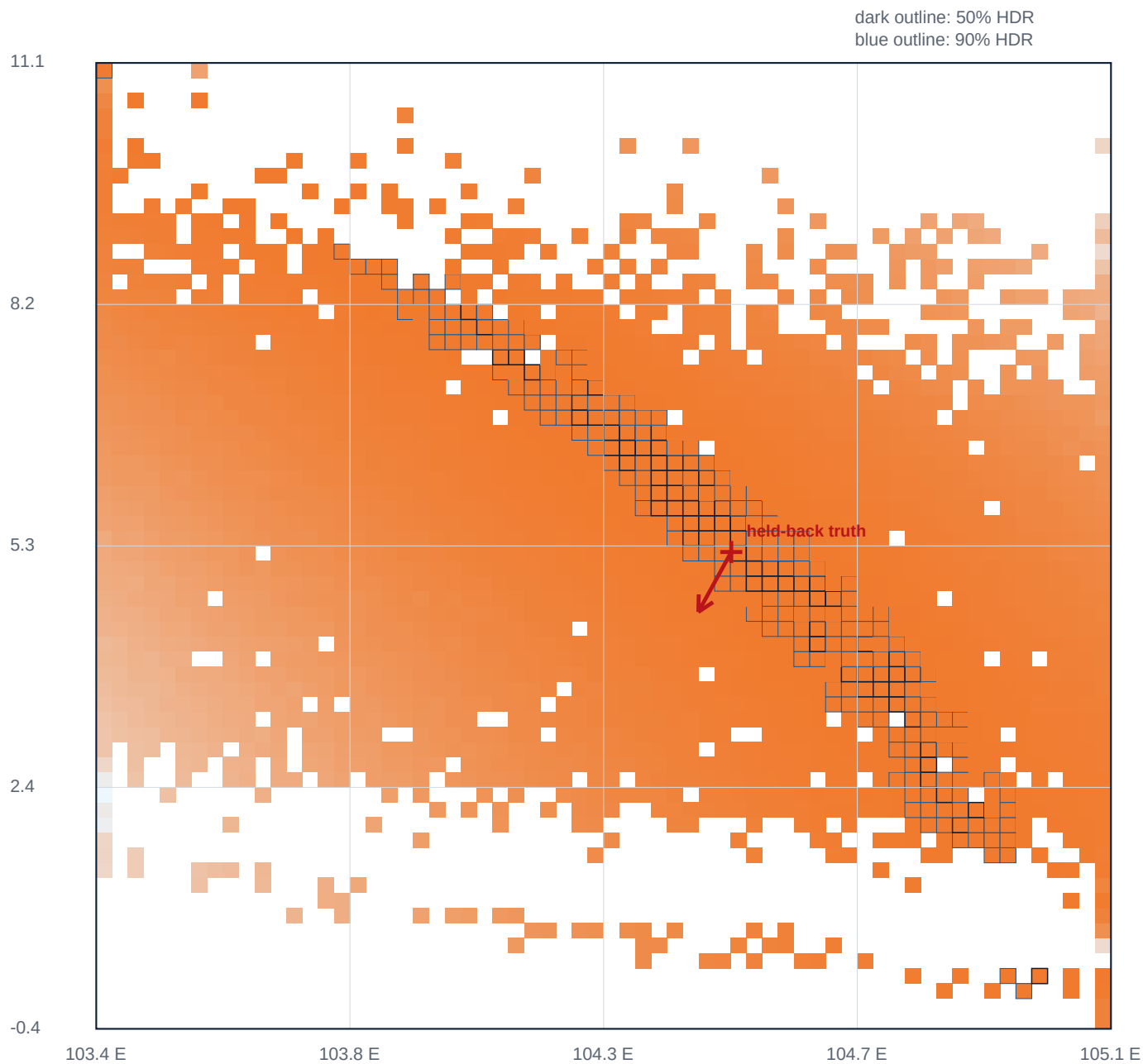
PARTICLE / ROOT ESS

222 / 34.9

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.

mh371_0648 SATCOM arc

2014-03-07T06:48:33.907Z | truth 5.2504 deg, 104.4874 deg | true heading 208.8 deg



Posterior and control metrics

MEAN POSITION ERROR

26.4 NM

MACH: POSTERIOR / TRUTH

0.787 / 0.821

GROUND SPEED: POSTERIOR / TRUTH

478 / 483 kt

TRUTH-IMPLIED BFO BIAS

168.6 Hz

POSTERIOR WITHIN 100 NM

60.4%

ALTITUDE: POSTERIOR / TRUTH

33995 / 40000 ft

LATENT BFO BIAS

169.1 Hz

PARTICLE / ROOT ESS

458 / 30.5

Density combines equal-weight deterministic seed replicates. The red cross and arrow were loaded only during scoring.